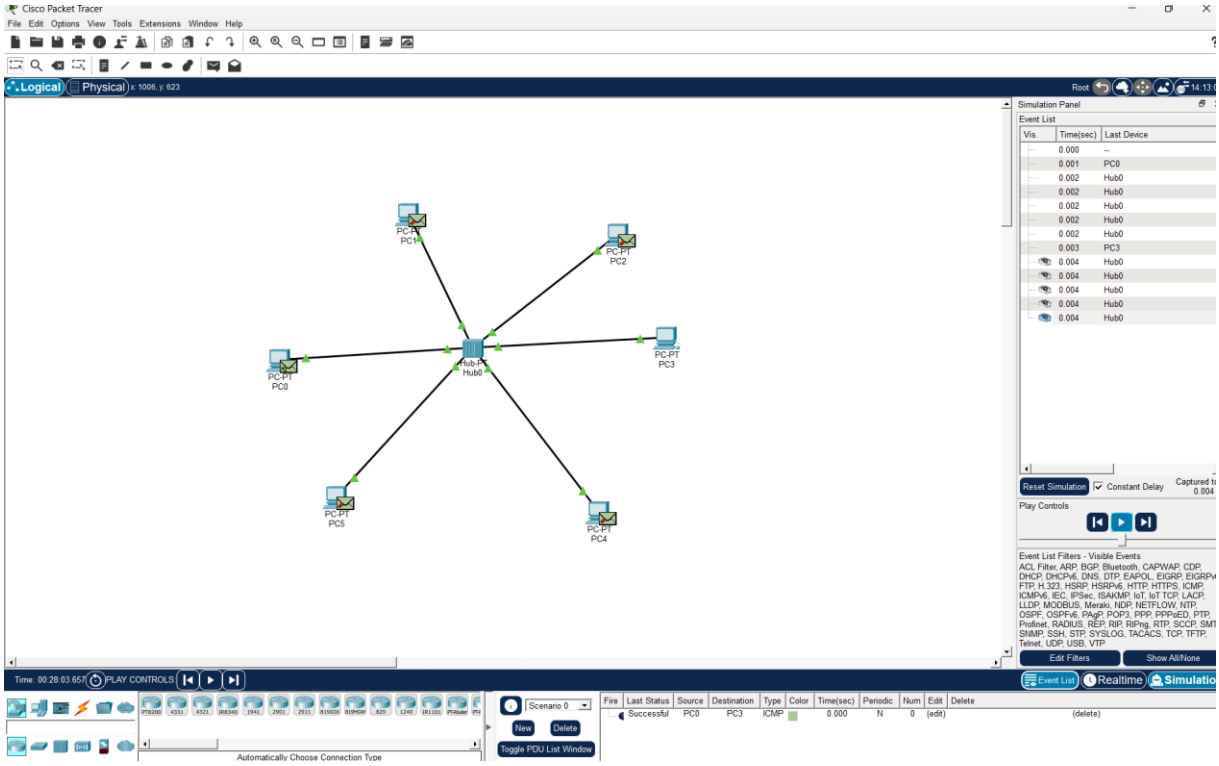
Name: Y. Varun Kumar Reddy

Reg No: 192424230

Course Name: Computer Networks for Software Application

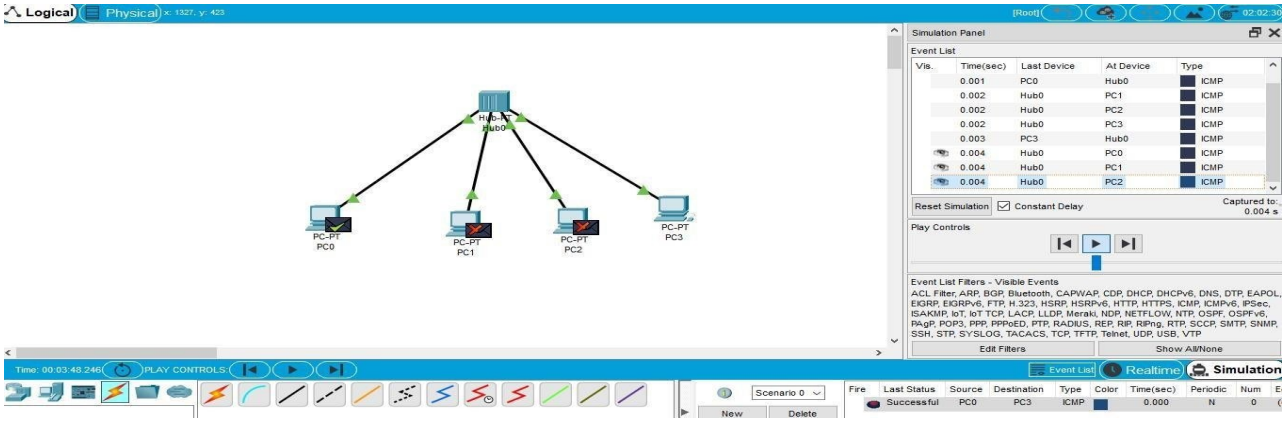
Code: CSA0713

1.Configuration of Network Devices using Packet Tracer tools (Hub, Switch, Ethernet, Broadcast)



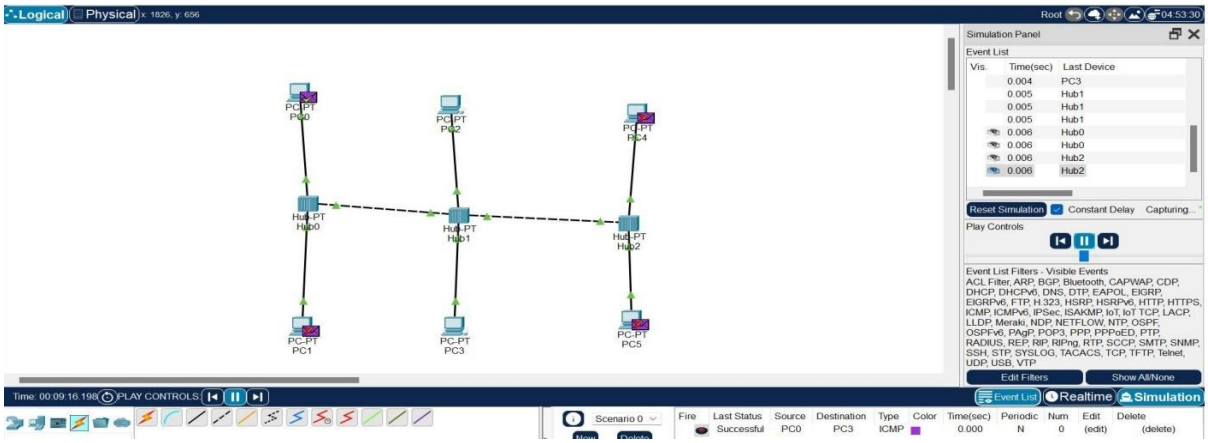
2. Design and Configuration of Star Topologies

using Packet Tracer Star Topology using Hub:



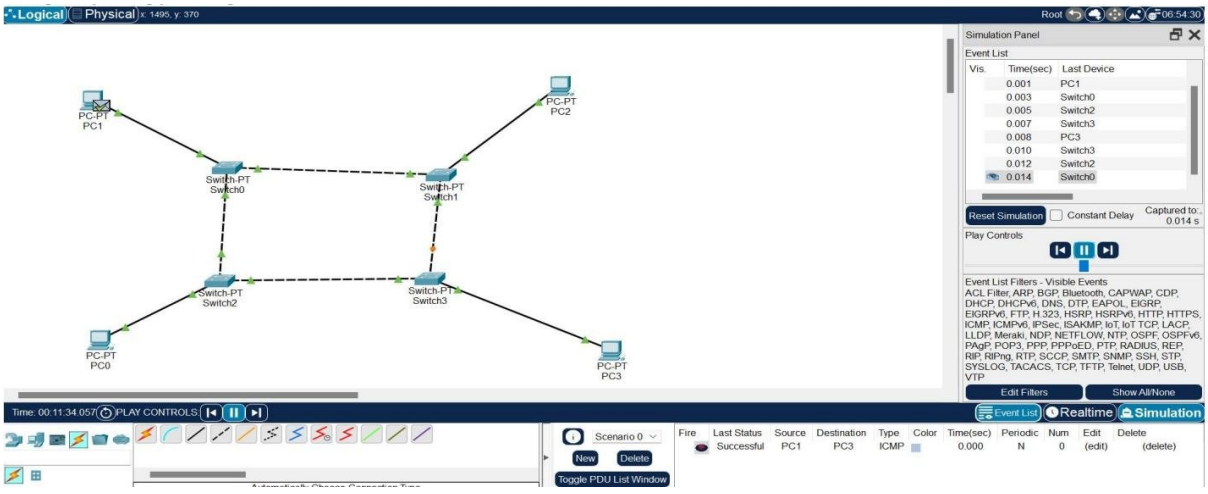
3. Design and Configuration of BUS Topologies
using Packet Tracer.

Bus Topology using Hub:



4. Design and Configuration of RING Topologies using Packet Tracer

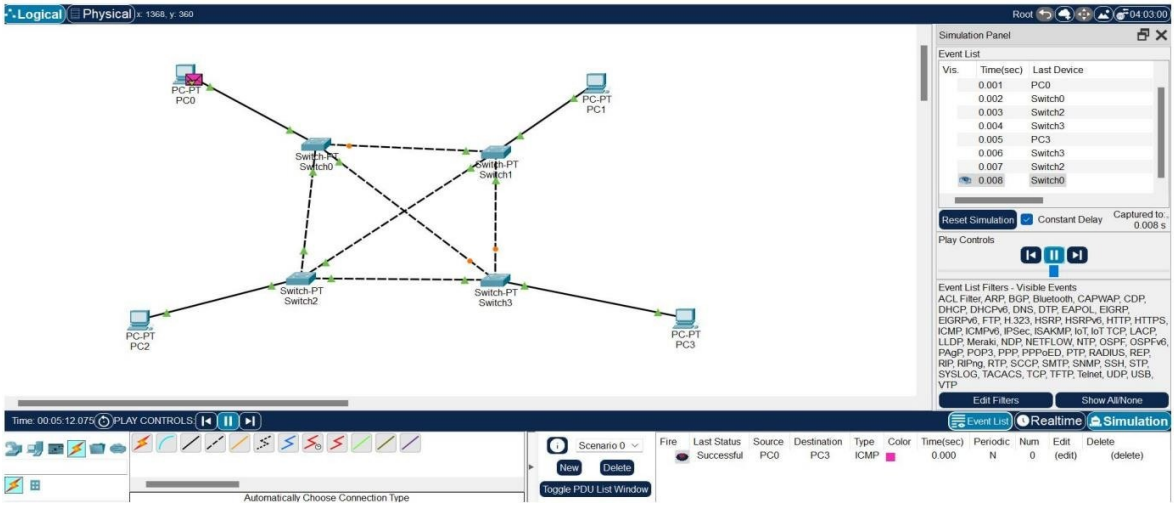
Ring Topology using Switch:



5. Design and Configuration of Mesh Topologies

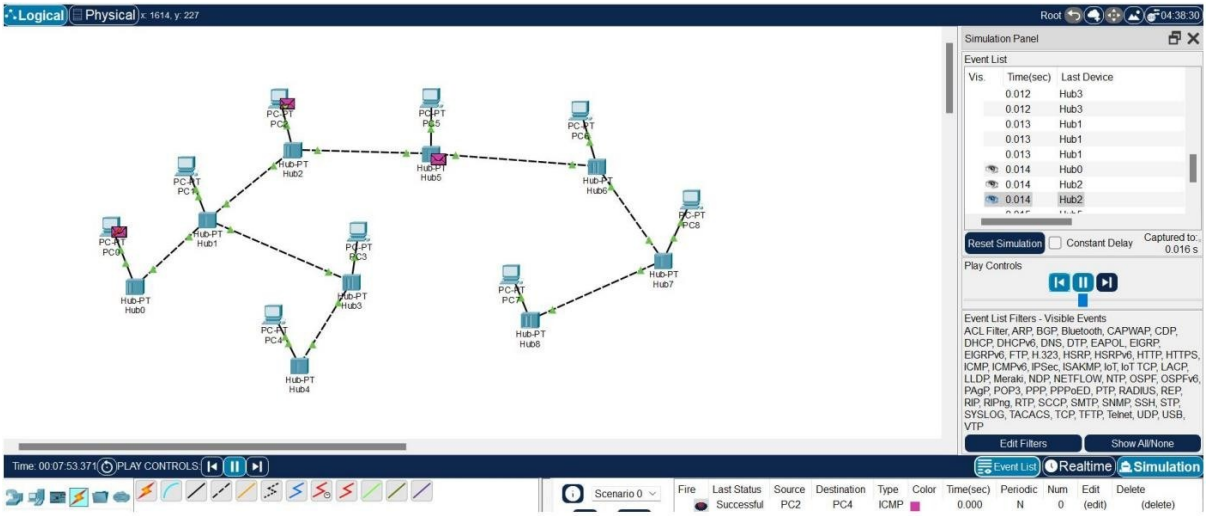
using Packet Tracer

Mesh Topology using Switch:



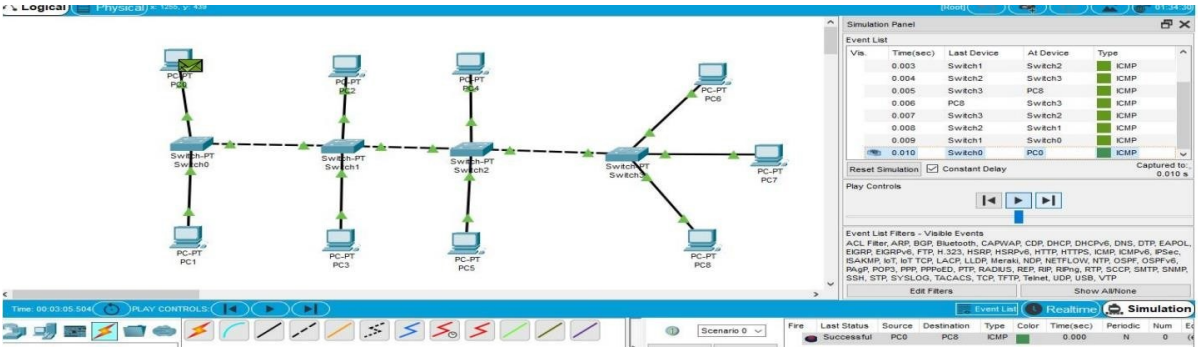
6.Design and Configuration of Tree Topologies using Packet Tracer.

Tree Topology using Hub:

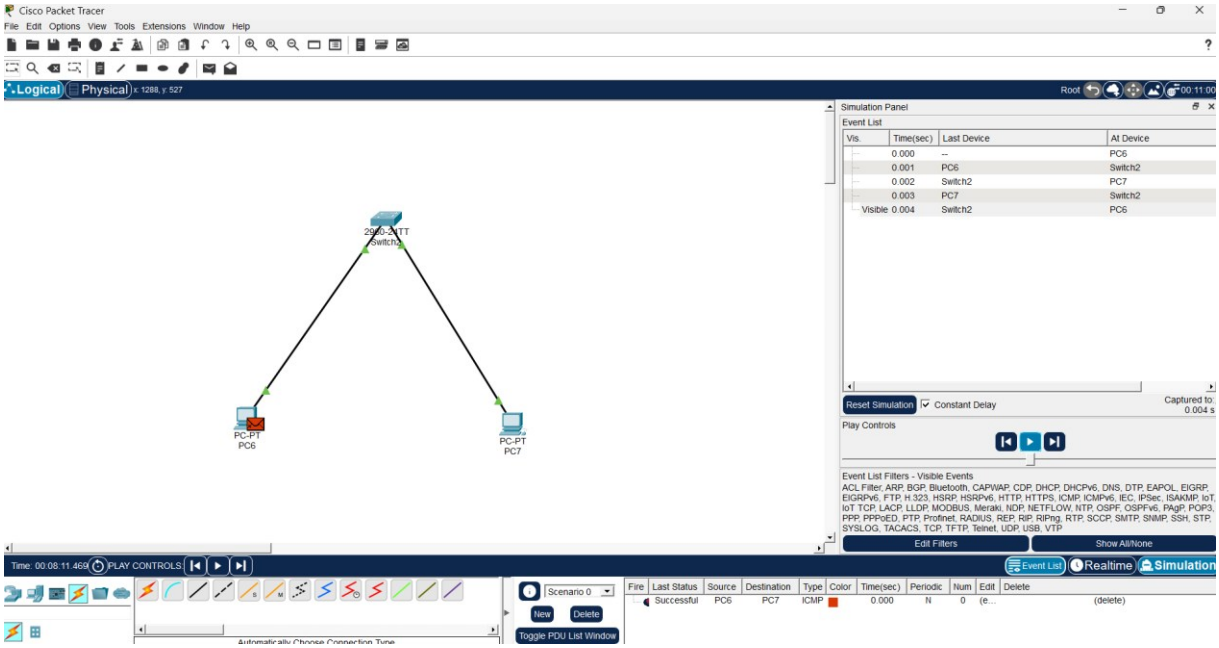


7.Design and Configuration of Hybrid Topologies using Packet Tracer.

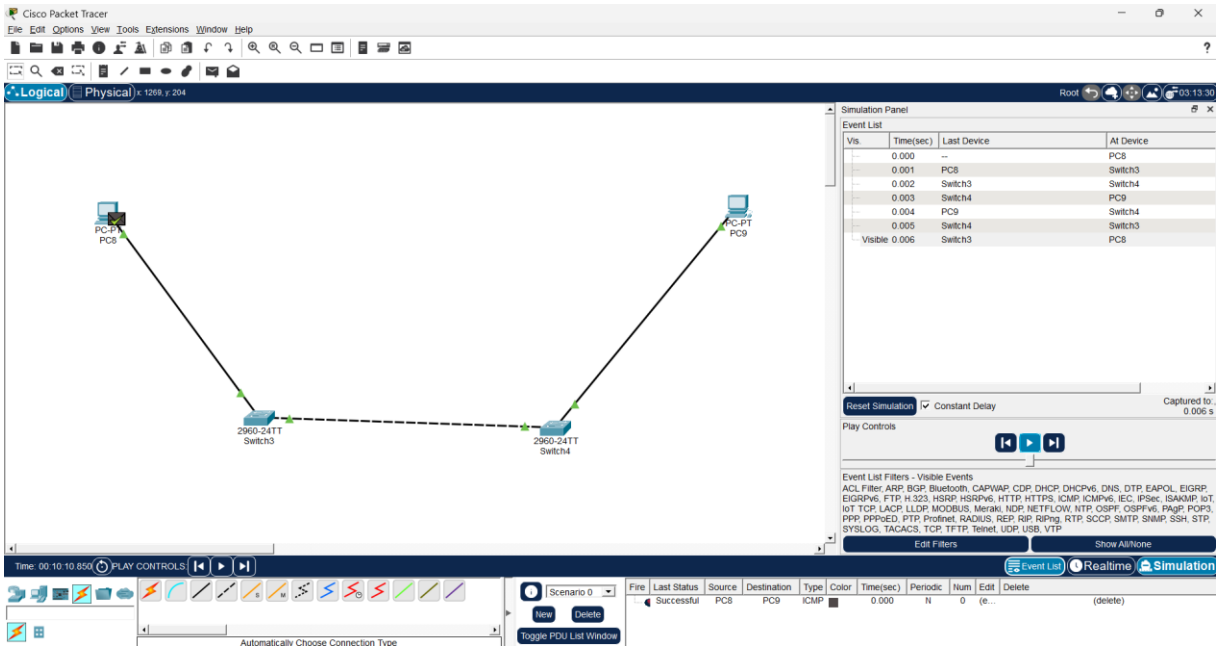
Hybrid Topology using Switch:



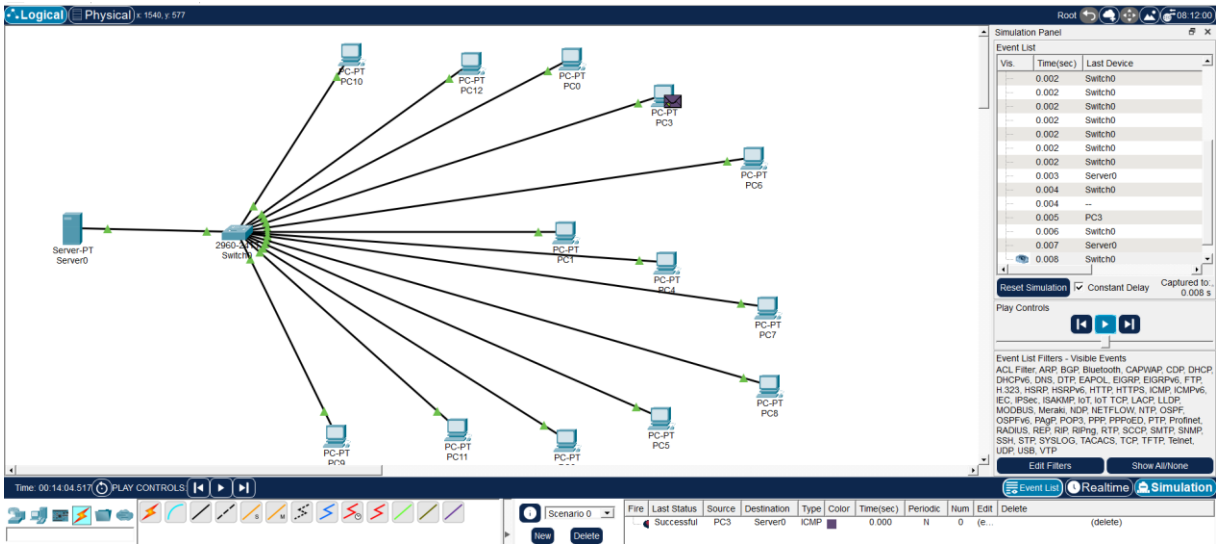
8.Data Link Layer Traffic Simulation using Packet Tracer Analysis of ARP.



9.Data Link Layer Traffic Simulation using Packet Tracer Analysis of LLDP.

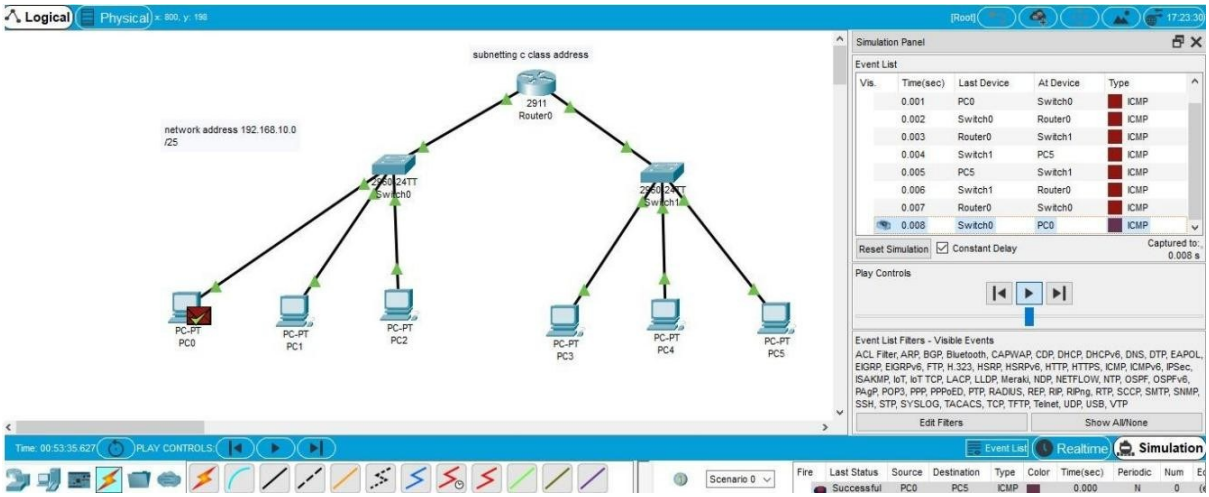


10.Making Computer Lab in Cisco Packet Tracer.

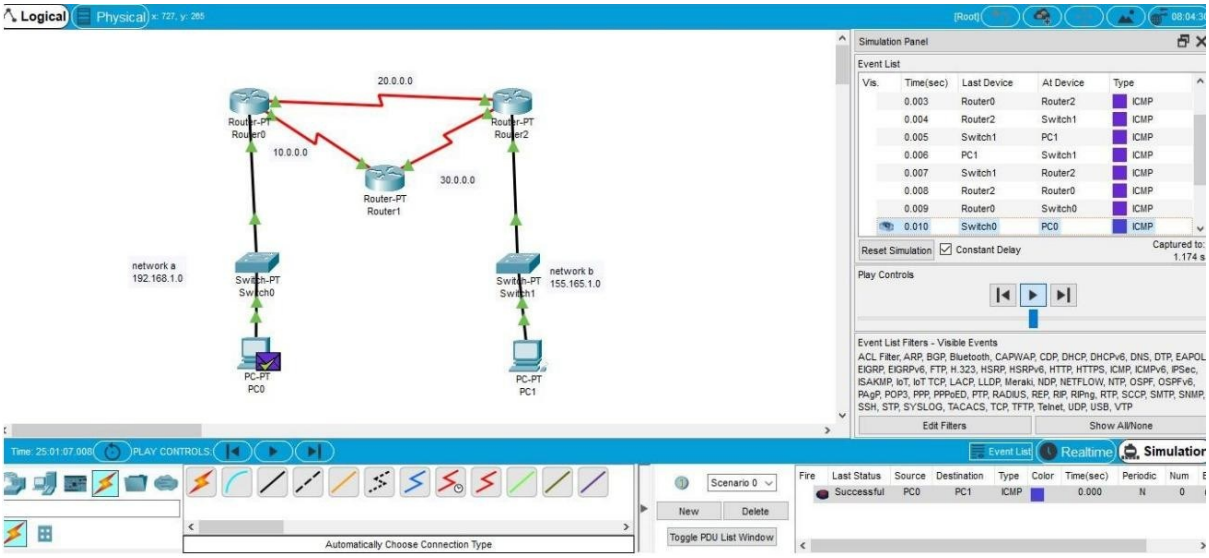


11.Designing two different network with Static Routing techniques using

Packet Tracer.

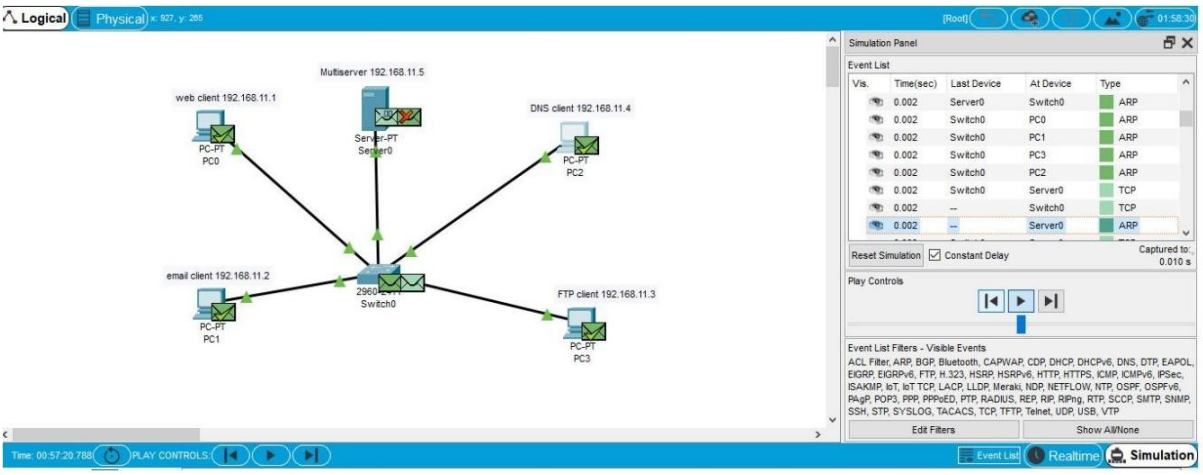


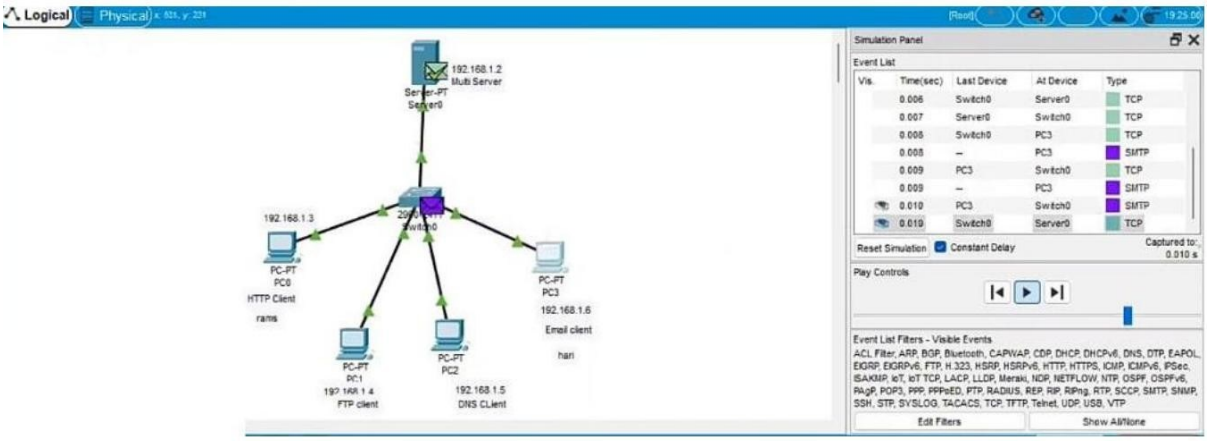
12. Design the Functionalities and Exploration of TCP using Packet Tracer.



13. Design the network model for Subnetting – Class C Addressing using

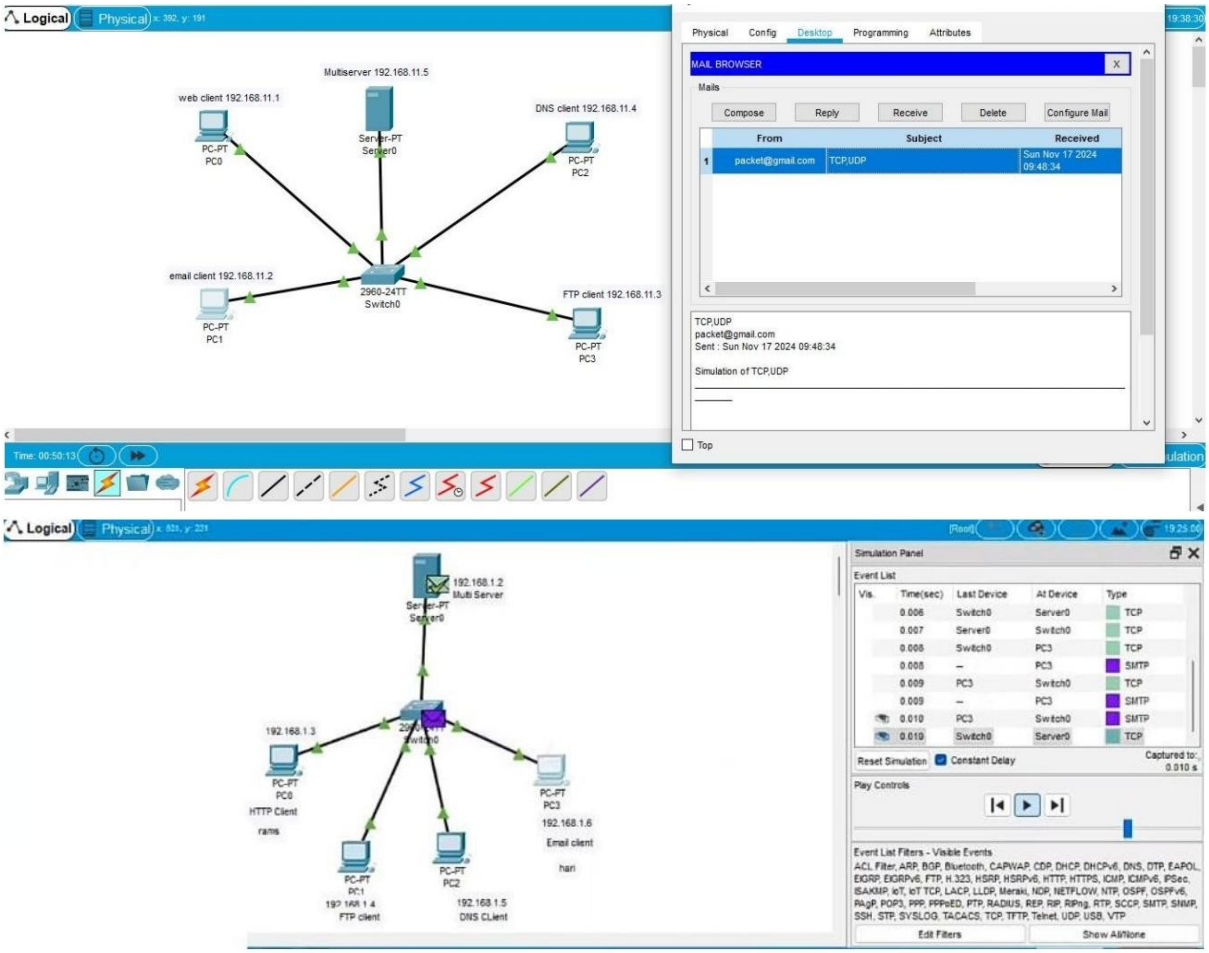
Packet Tracer.



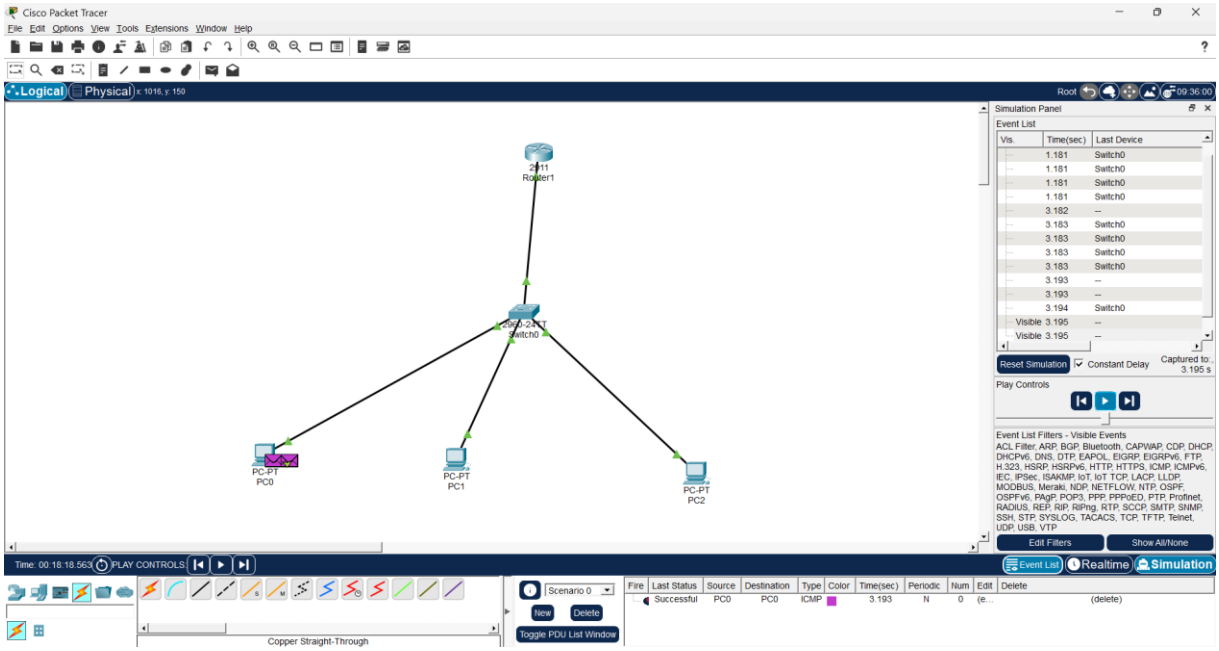


14. Simulating X, Y, Z Company Network Design and simulate using Packet

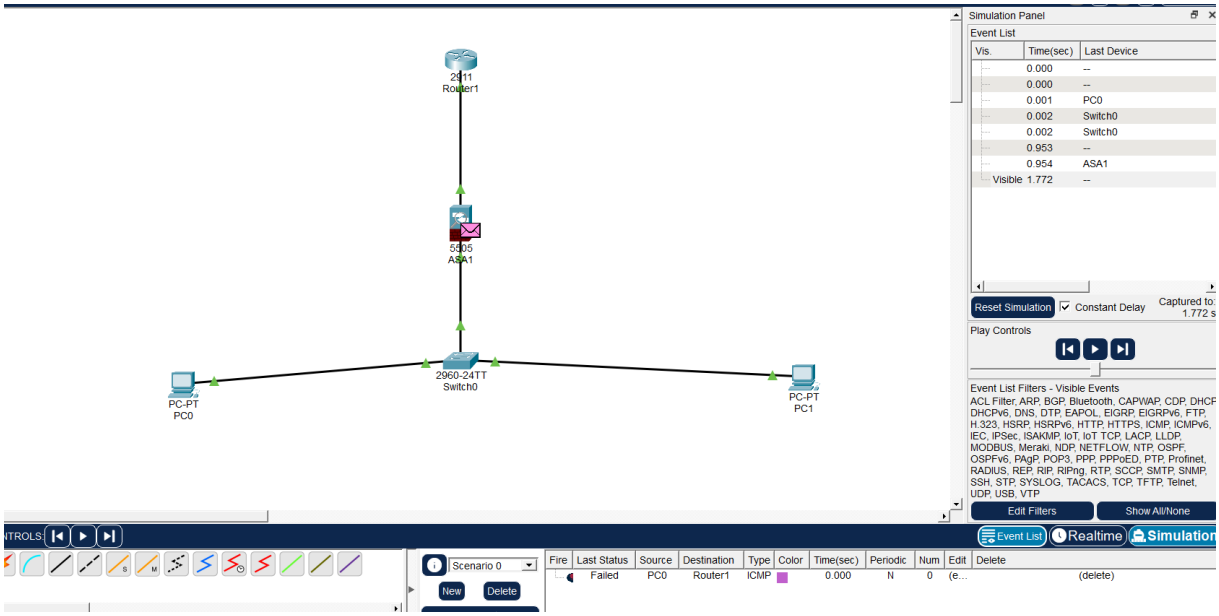
Tracer.



15. Configuration of DHCP (dynamic host configuration protocol) in packet Tracer.

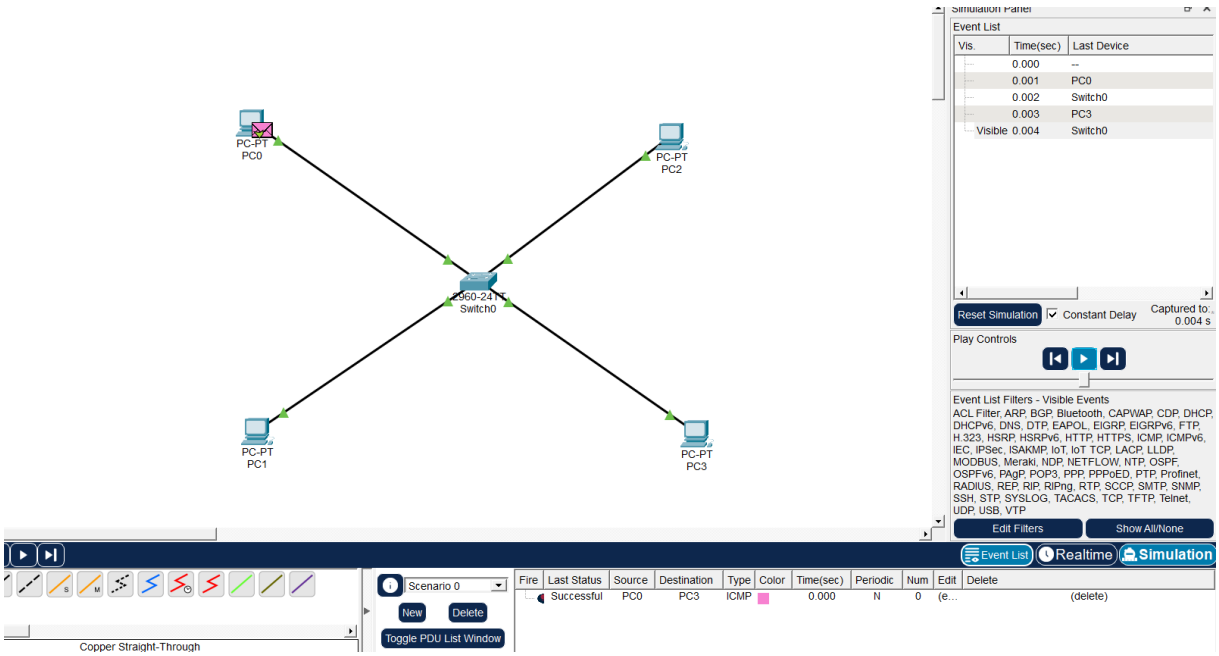


16. Configuration of firewall in packet tracer.

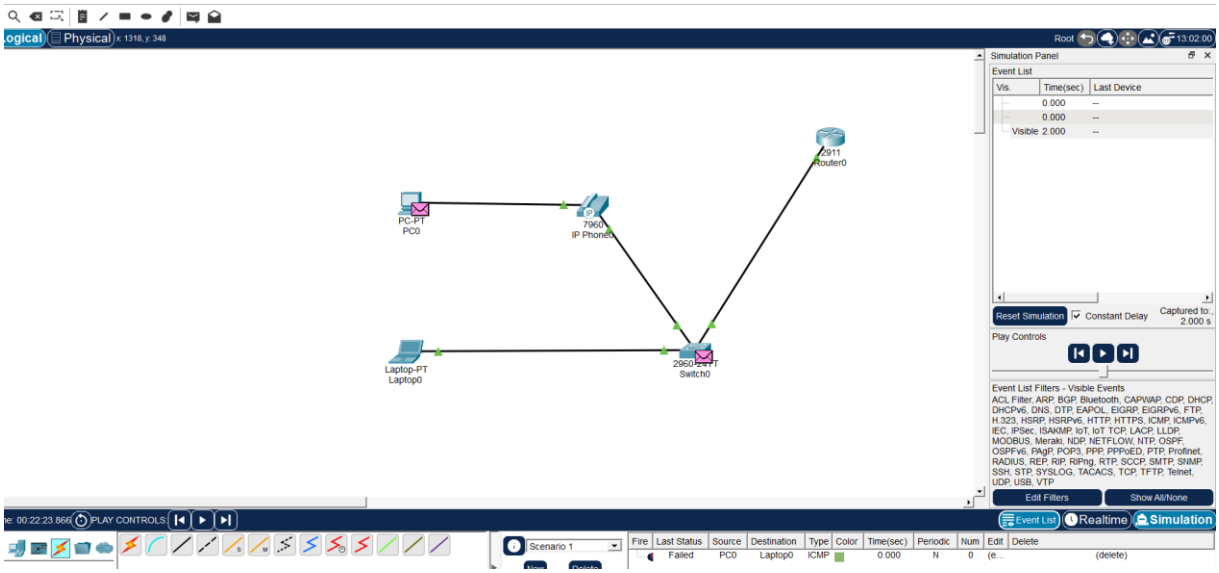


17. Make a Computer Lab to transfer a message from one node to another to

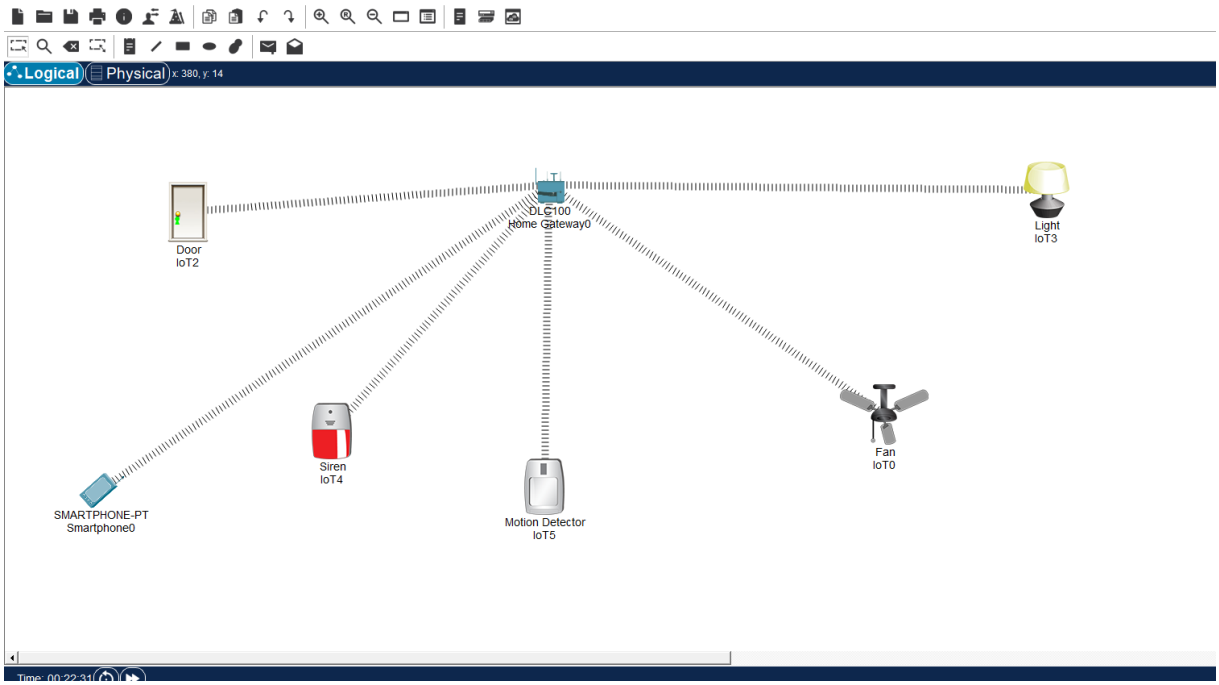
design and simulate using Cisco Packet Tracer.



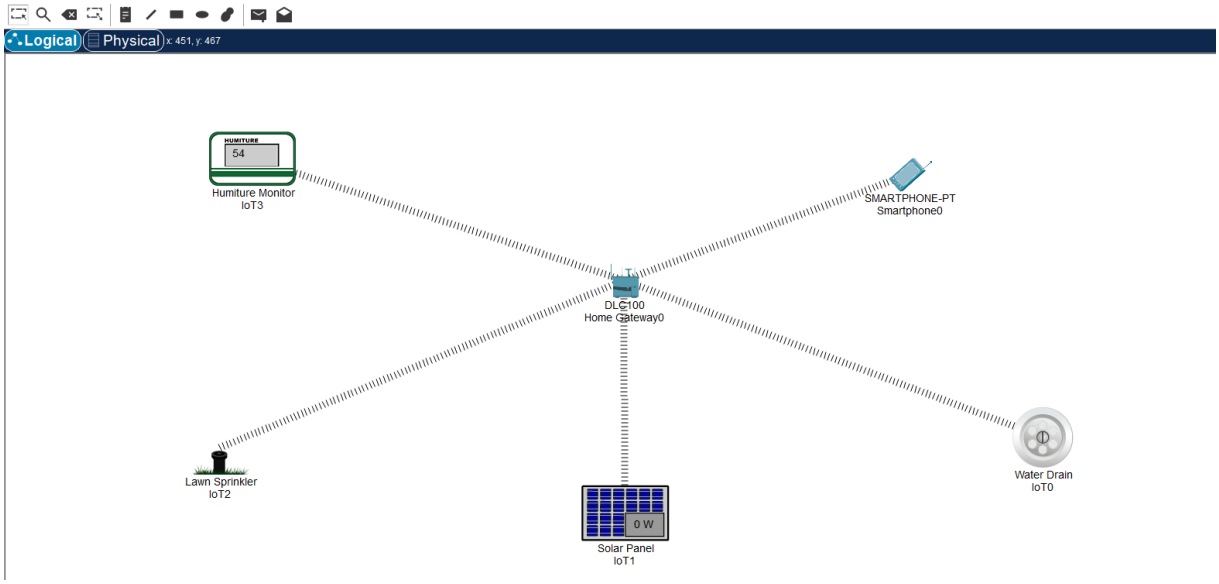
18. Simulate a Multimedia Network in Cisco Packet Tracer.



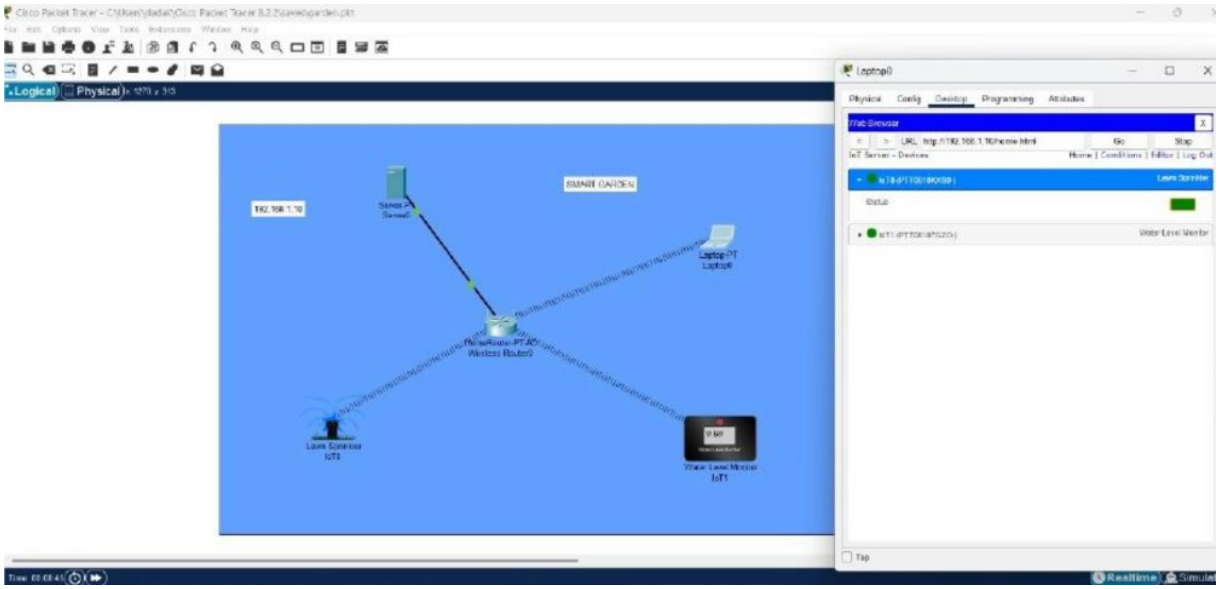
19. IoT based smart home applications. CO3



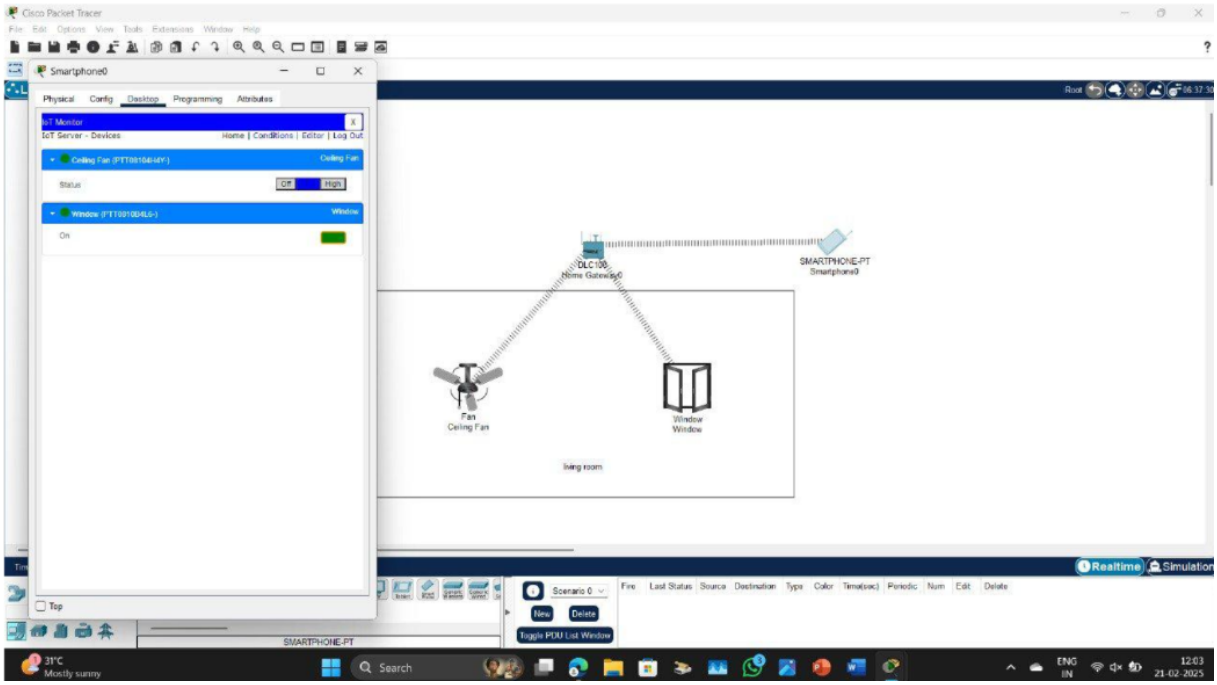
20. Implementation of IoT based smart gardening. CO2



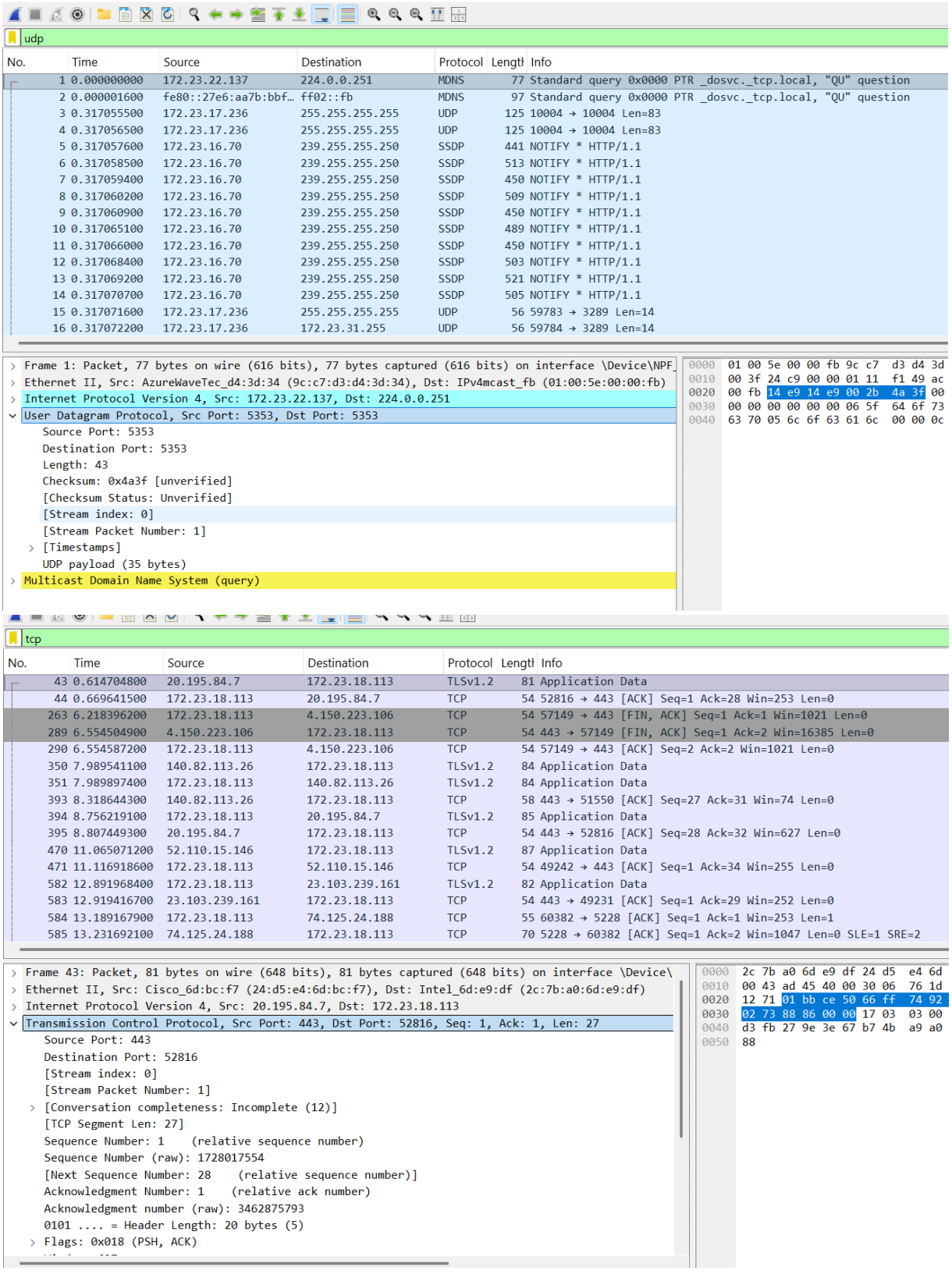
21. Implementation of IoT devices in networking. CO4



22. IoT based AAA Local and Server based authentication configuration. CO4

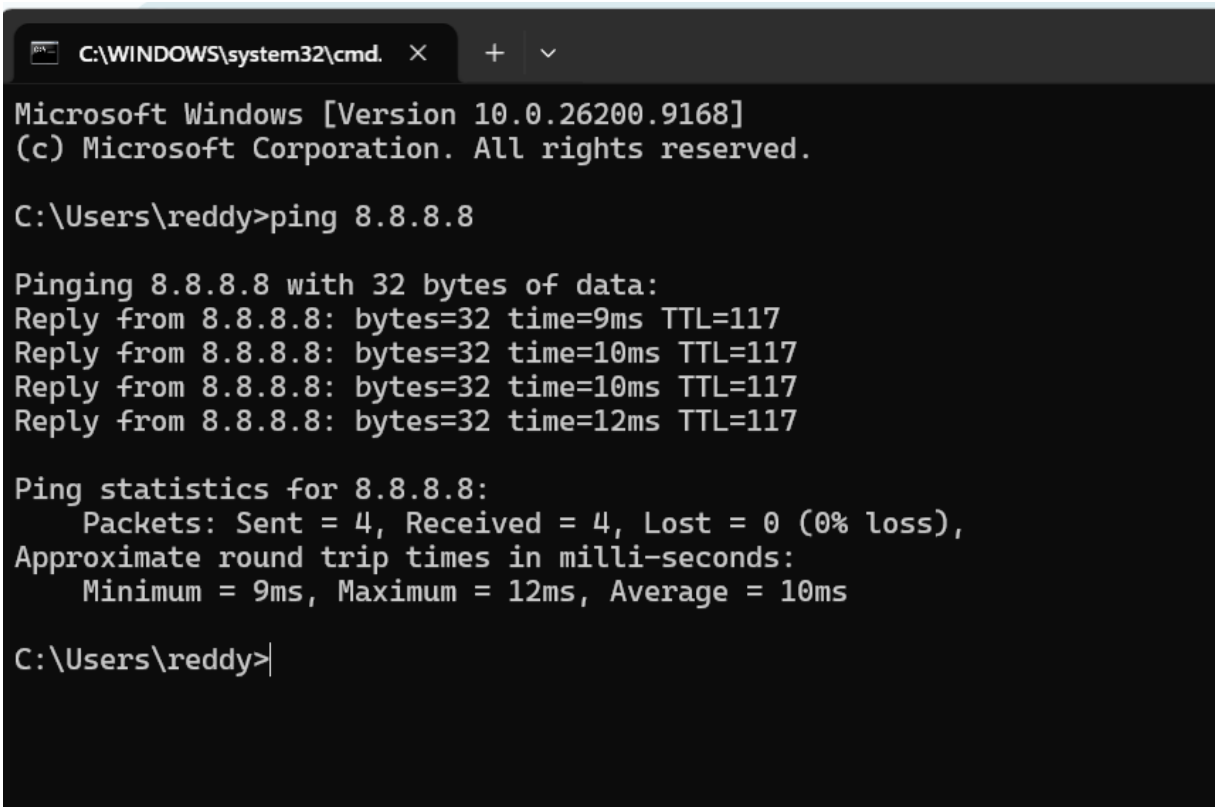


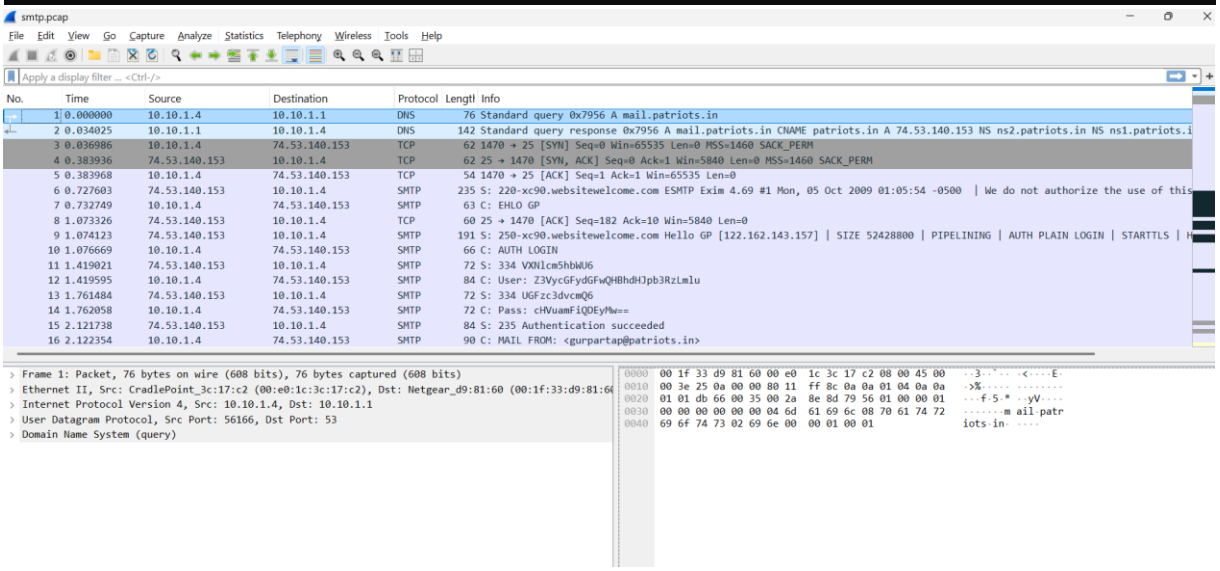
23. Transport layer protocol header analysis using Wire shark- TCP and UDP. CO5



24. Network layer protocol header analysis using Wire shark – SMTP and

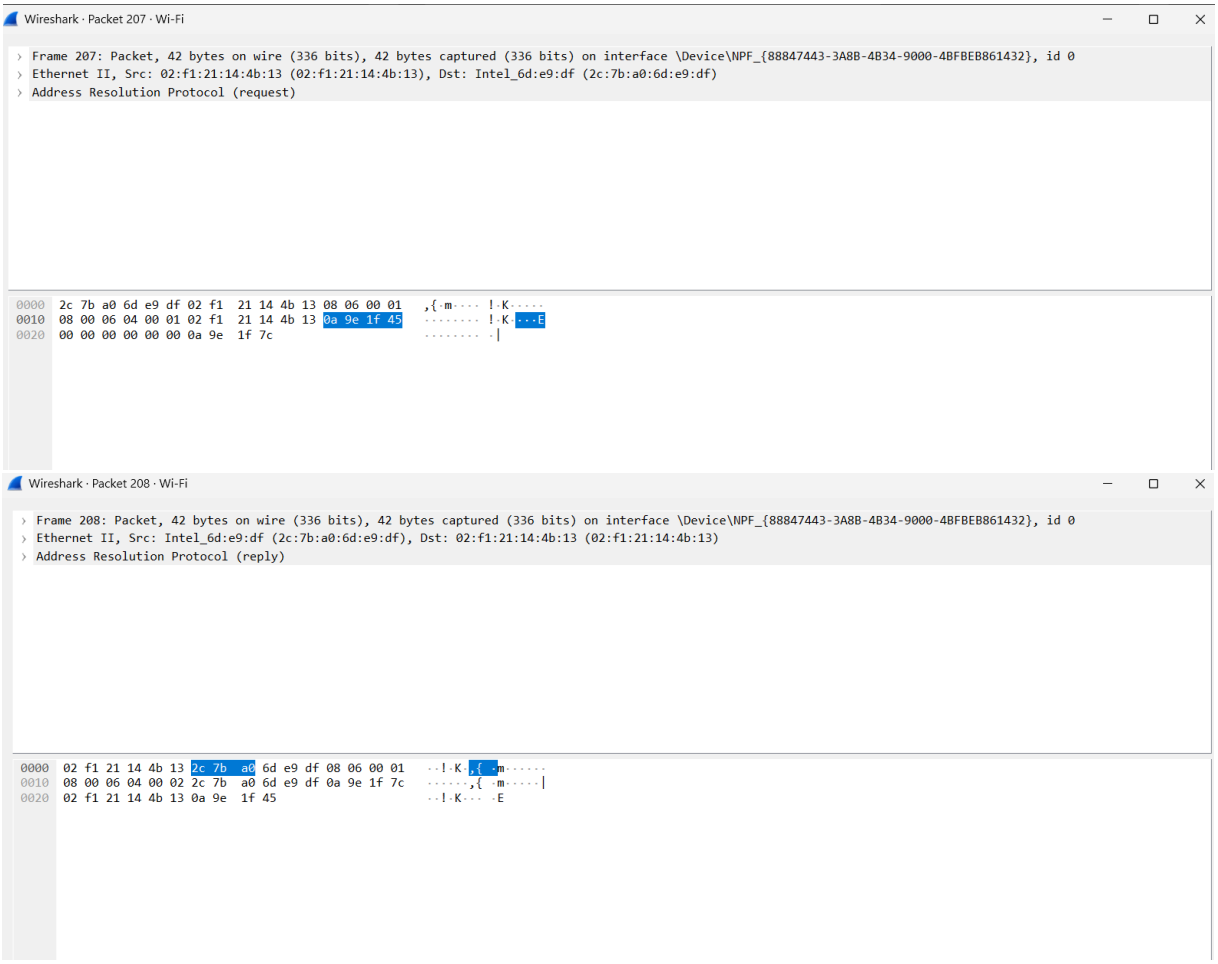
ICMP.


C:\WINDOWS\system32\cmd.
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.
C:\Users\reddy>ping 8.8.8.8
Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=9ms TTL=117
Reply from 8.8.8.8: bytes=32 time=10ms TTL=117
Reply from 8.8.8.8: bytes=32 time=10ms TTL=117
Reply from 8.8.8.8: bytes=32 time=12ms TTL=117
Ping statistics for 8.8.8.8:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 9ms, Maximum = 12ms, Average = 10ms
C:\Users\reddy>

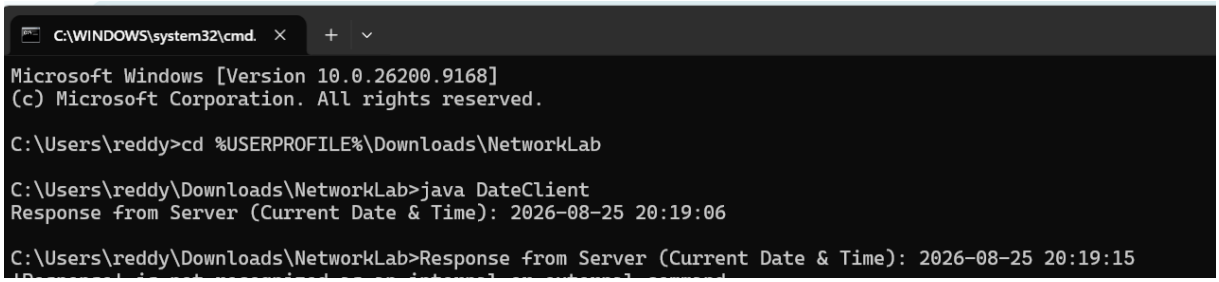

smtp.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info
1 0.000000 10.10.1.4 10.10.1.1 DNS 76 Standard query 0x7956 A mail.patriots.in
2 0.034025 10.10.1.1 10.10.1.4 DNS 142 Standard query response 0x7956 A mail.patriots.in CNAME patriots.in A 74.53.140.153 NS ns2.patriots.in NS ns1.patriots.i
3 0.036986 10.10.1.4 74.53.140.153 TCP 62 1470 → 25 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM
4 0.383936 74.53.140.153 10.10.1.4 TCP 62 25 → 1470 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460 SACK_PERM
5 0.383968 10.10.1.4 74.53.140.153 TCP 54 1470 → 25 [ACK] Seq=1 Ack=1 Win=65535 Len=0
6 0.727603 74.53.140.153 10.10.1.4 SMTP 235 S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 | We do not authorize the use of this
7 0.732749 10.10.1.4 74.53.140.153 SMTP 63 C: EHLO GP
8 1.073326 74.53.140.153 10.10.1.4 TCP 60 25 → 1470 [ACK] Seq=182 Ack=10 Win=5840 Len=0
9 1.074123 74.53.140.153 10.10.1.4 SMTP 191 S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] | SIZE 52428800 | PIPELINING | AUTH PLAIN LOGIN | STARTTLS |
10 1.076669 10.10.1.4 74.53.140.153 SMTP 66 C: AUTH LOGIN
11 1.419021 74.53.140.153 10.10.1.4 SMTP 72 S: 334 VXNlcm5hbWU6
12 1.419595 10.10.1.4 74.53.140.153 SMTP 84 C: User: Z3VycGFydGFwQHBhdHJpb3RzLmlu
13 1.761484 74.53.140.153 10.10.1.4 SMTP 72 S: 334 UGFzc3dvcmQ6
14 1.762058 10.10.1.4 74.53.140.153 SMTP 72 C: Pass: cHVuamFiQDEyMw==
15 2.121738 74.53.140.153 10.10.1.4 SMTP 84 S: 235 Authentication succeeded
16 2.122354 10.10.1.4 74.53.140.153 SMTP 90 C: MAIL FROM: <gurpartap@patriots.in>
Frame 1: Packet, 76 bytes on wire (608 bits), 76 bytes captured (608 bits)
Ethernet II, Src: CradlePoint_3c:17:c2 (00:e0:1c:3c:17:c2), Dst: Netgear_d9:81:60 (00:1f:33:d9:81:6
Internet Protocol Version 4, Src: 10.10.1.4, Dst: 10.10.1.1
User Datagram Protocol, Src Port: 56166, Dst Port: 53
Domain Name System (query)
0000 00 1f 33 d9 81 60 00 e0 1c 3c 17 c2 08 00 45 00
0010 00 3e 25 0a 00 00 80 11 ff 8c 0a 0a 01 04 0a 0a
0020 01 01 db 66 00 35 00 2a 8e 8d 79 56 01 00 00 01
0030 00 00 00 00 00 00 04 6d 61 69 6c 08 70 61 74 72
0040 69 6f 74 73 02 69 6e 00 00 01 00 01

CO4

25. Network layer protocol header analysis using Wire shark – ARP and HTTP.

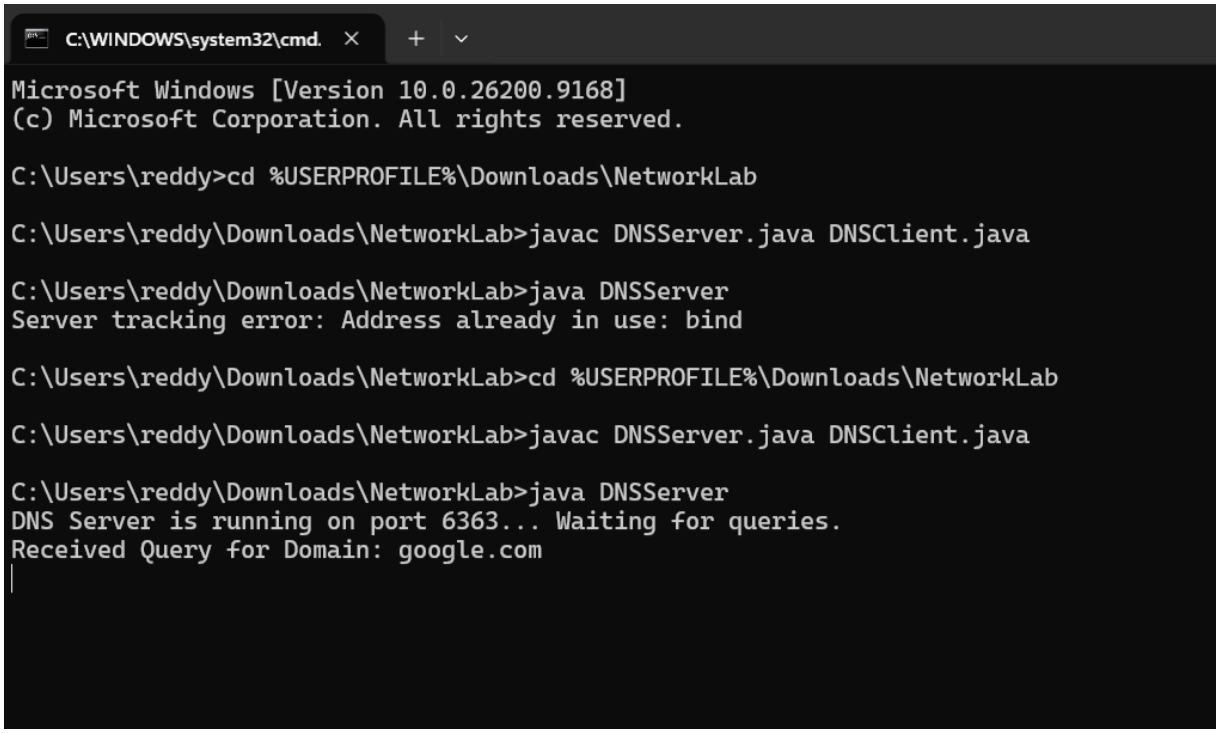


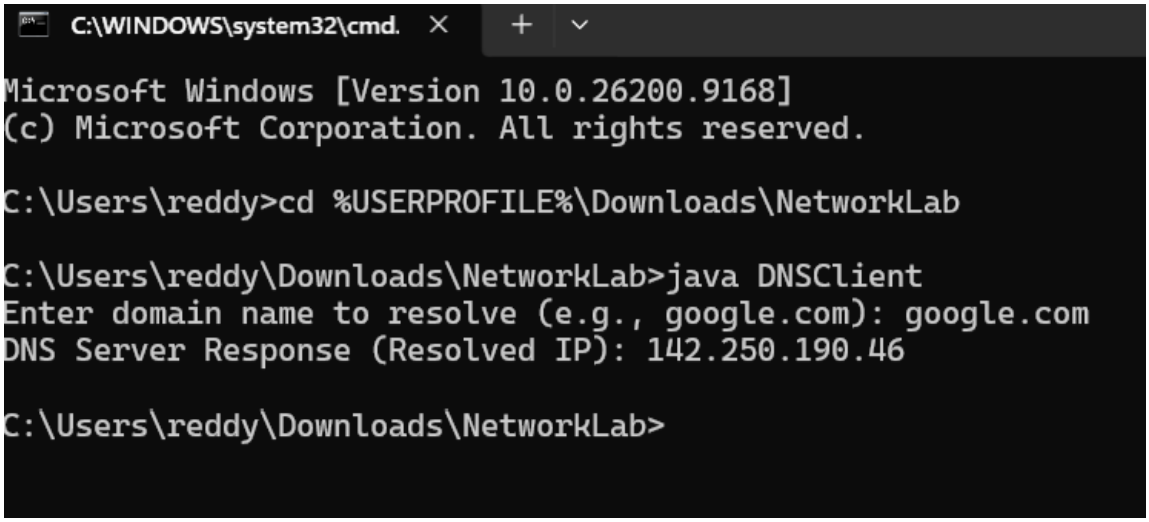
26. Implementation of date and time display from client to server using TCP
sockets in java/C.



CO2

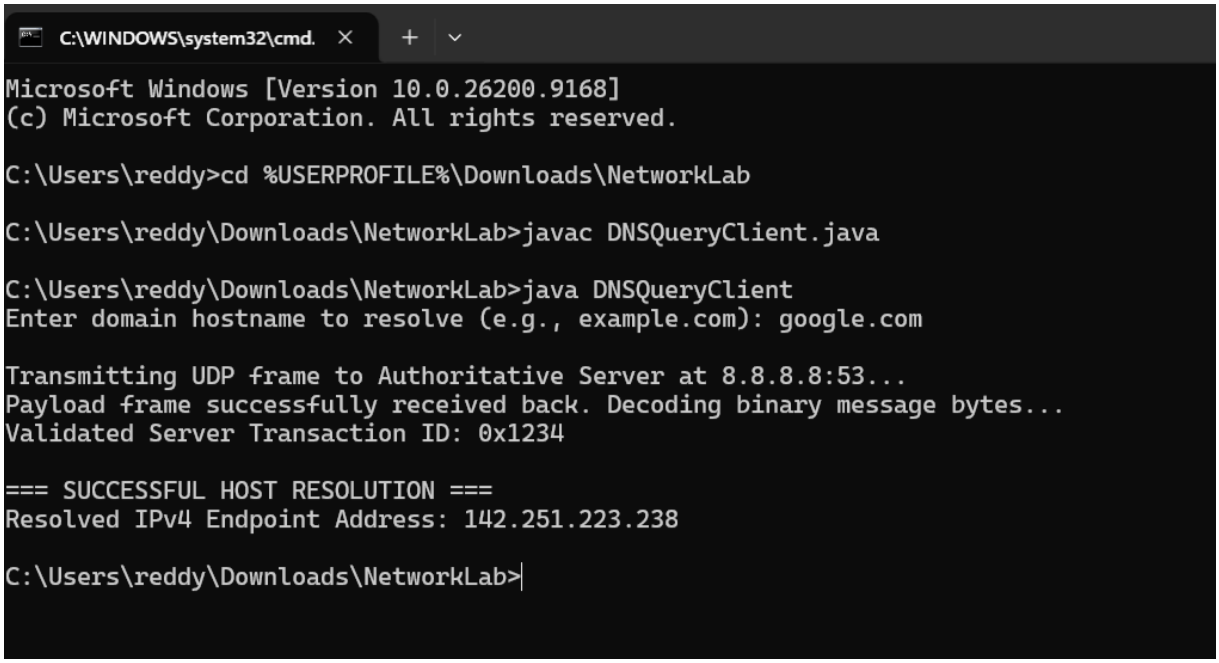
27. Implementation of a DNS server and client in java/C using UDP sockets. CO3



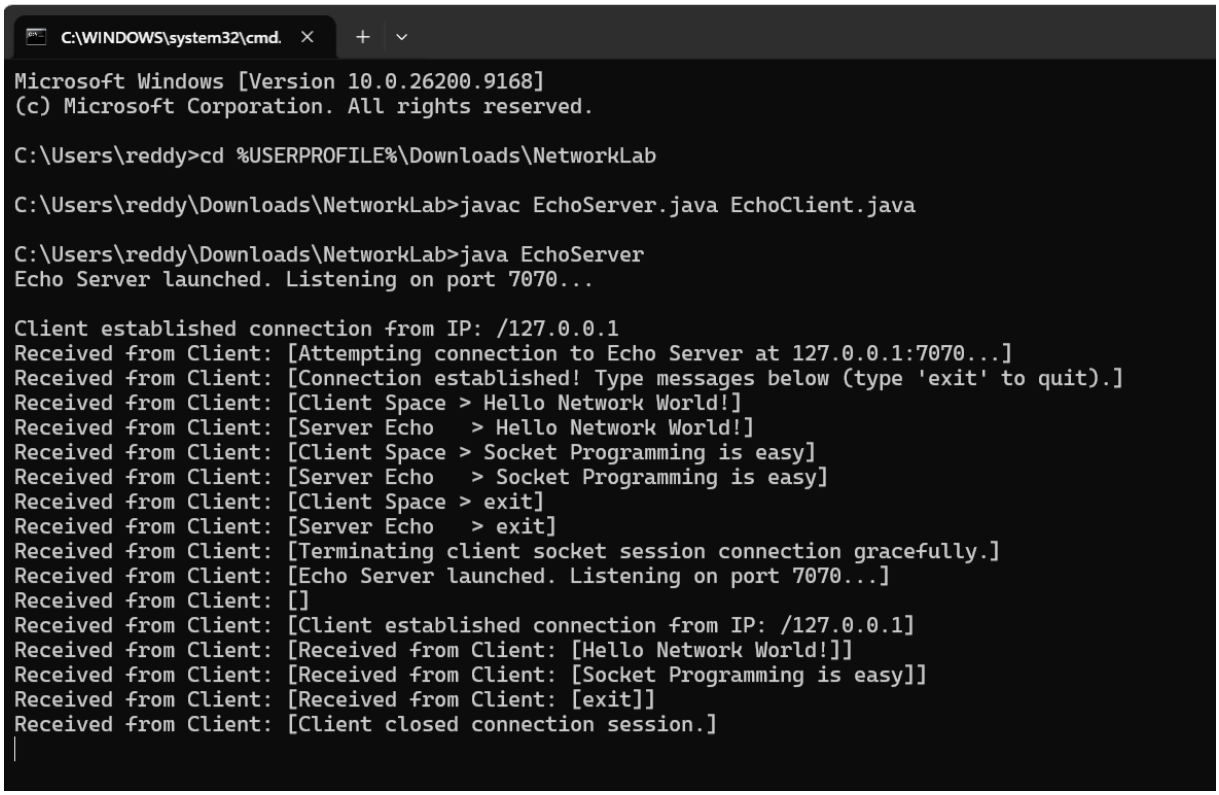


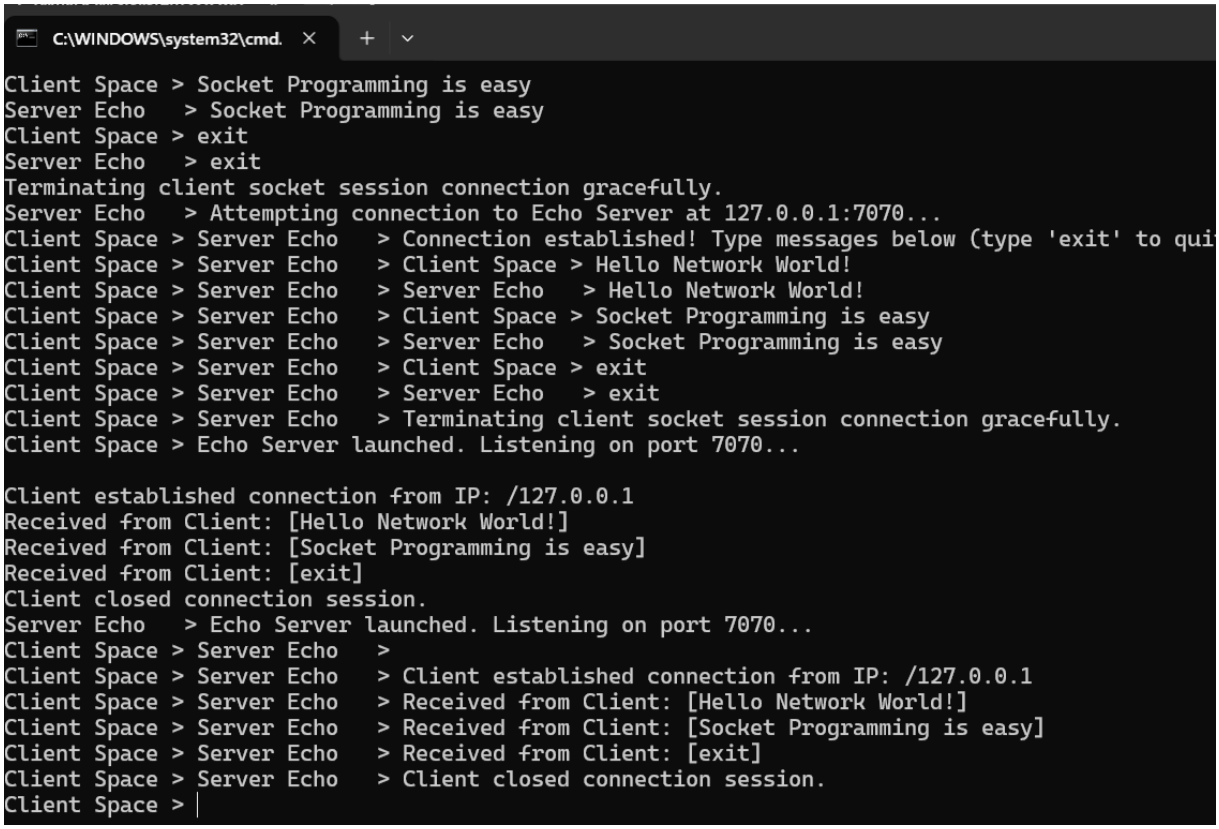
28. Developing a client that contacts a given DNS server to resolve a given
hostname in java/C.

CO4

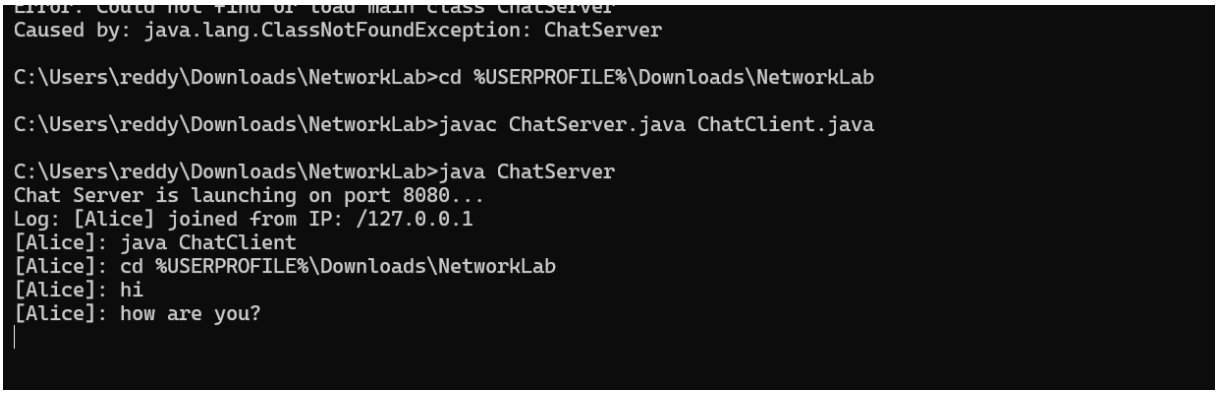


29. Creating the applications using TCP echo server and client in java/C. CO4





30. Creating the applications using TCP chat client and chat server in java/C. CO3



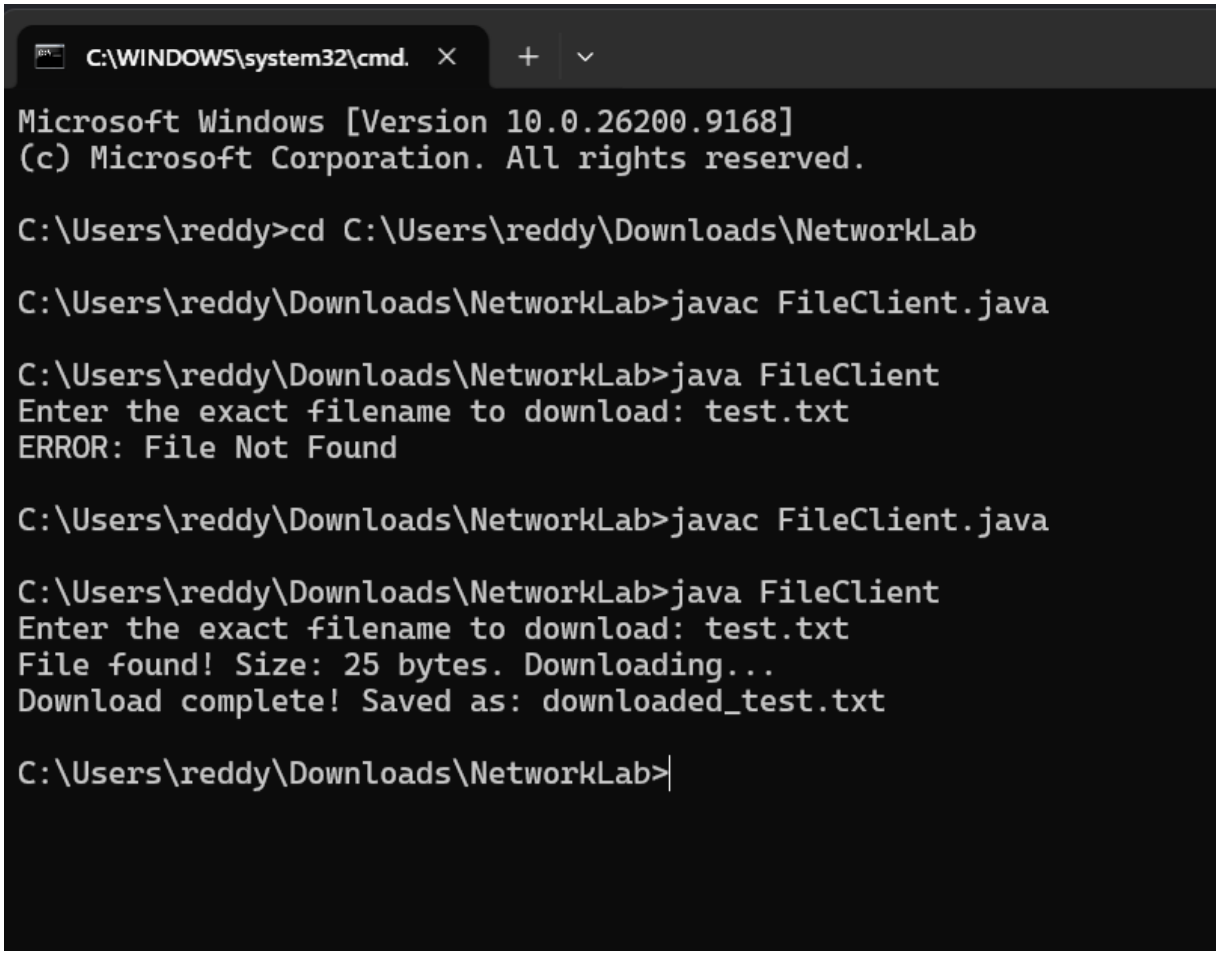
31. Implementing ARP protocols in java/C. CO5

```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\reddy>cd Downloads\NetworkLab

C:\Users\reddy\Downloads\NetworkLab>javac ARPClient.java

C:\Users\reddy\Downloads\NetworkLab>java ARPClient
Enter IP Address: 10.10.10.1
The corresponding MAC Address is: MAC Address Not Found

C:\Users\reddy\Downloads\NetworkLab>java ARPClient
Enter IP Address: 192.168.1.10
The corresponding MAC Address is: 00-11-22-33-44-55

C:\Users\reddy\Downloads\NetworkLab>
```

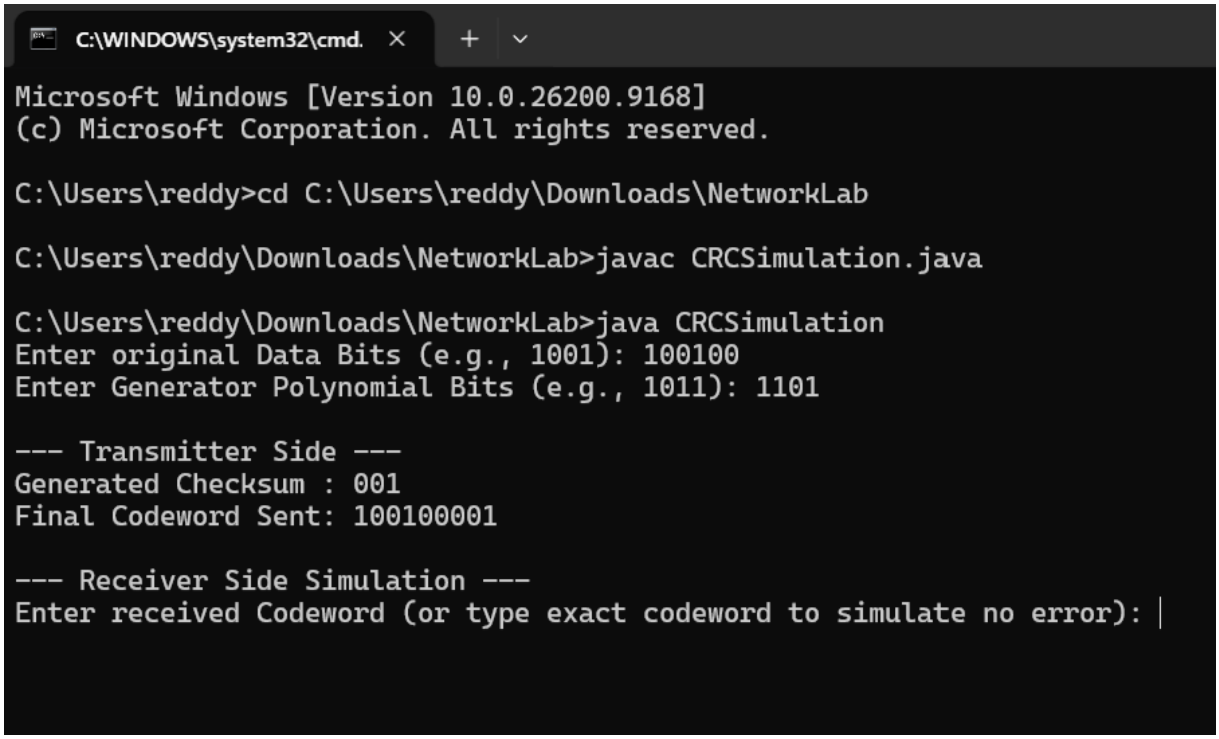
32. Implementation of Bit stuffing mechanism using C. CO5

```
C:\WINDOWS\system32\cmd.

(c) Microsoft Corporation. All rights reserved.

C:\Users\reddy>cd C:\Users\reddy\Downloads\NetworkLab

C:\Users\reddy\Downloads\NetworkLab>javac BitStuffing.java
BitStuffing.java:52: error: cannot find symbol
        Scanner scanner = new Scanner(System.?n == null ? System.in : System.in);
                                              ^
  symbol:   variable ?n
  location: class System
1 error

C:\Users\reddy\Downloads\NetworkLab>java BitStuffing
Error: Could not find or load main class BitStuffing
Caused by: java.lang.ClassNotFoundException: BitStuffing

C:\Users\reddy\Downloads\NetworkLab>javac BitStuffing.java

C:\Users\reddy\Downloads\NetworkLab>java BitStuffing
Enter the binary data (0s and 1s): 1010

--- Transmitter Side ---
Original Data: 1010
Stuffed Data : 01111110 1010 01111110 (with flags)

--- Receiver Side ---
Destuffed Data: 1010

C:\Users\reddy\Downloads\NetworkLab>
```

33. Implementing the applications using TCP file transfer in java/C. CO5



34. Implementing the simulation of error correction code - CRC in java/C. CO5



35. Implementing the sliding window protocol in java/C.